

Hello, I am

# Jang Byeong Heon

**An AI service developer who builds fast with AI coding agents and verifies with reproduction tests and measurements.**

I separate what the LLM decides from what code decides, write a reproduction test before each fix, and report results as before/after numbers with their measurement conditions. When AI wrote the code, I say so. A Philosophy degree in problem structuring and AI/SW training at SSAFY (1,628 hours) and SKALA Cohort 4 are how I explain why things were built the way they were.

## 01

Ops Sentinel — fixed a concurrency defect that had been masked by growing the connection pool, after isolating its causes with reproduction tests. Pool 10, 150 concurrent requests to one resource, 3 s AI delay: 500 errors 372 → 0, audit logs 450/450 (re-measured Sep 2026, H2 in-memory)

## 02

RE:RUN — tested the 'AI fixes itself' claim with a pre-registered experiment. Removing the hint from the diagnosis prompt dropped correct self-repair from 72% to 0% (n=25 per condition), and I put that result on the README's first screen

## 03

JangBogo — a deterministic policy engine approves payments; the LLM only writes the rationale. Fixed a payment race found in a post-hoc check: max orders from one concurrently resent payment proof 10 → 1 (650 ms injected verification delay, up to 10 concurrent × 5 runs)

My background in Philosophy trained me to analyze the conditions of human judgment, choice, and trust. This allows me to look past surface-level requirements to understand 'why' users behave a certain way. I have evolved into a PM who interprets problems deeply and realizes those interpretations as functioning systems.

I learned Python, algorithms, web, and AI fundamentals at SSAFY, and backend and AI service design at SKALA Cohort 4. Now I hand implementation to AI coding agents, but I set acceptance criteria first, verify the output with separate review and reproduction tests, and re-measure every number I claim. I see technical understanding not as something to show off but as the way to check what is actually true.

"I treat AI as a layer for better decisions, not as a gimmick." I see AI not as a simple automation tool, but as a support layer that helps users think and decide more clearly. In the DailyLog project, I solved the 'blank screen' problem with an AI-driven questioning flow, turning technical elements into real service value.

### Core Competencies

- LLM Trust Boundary Design
- Reproduce-First Debugging
- Before/After Measurement
- Agent-Built, Human-Verified

AT A GLANCE

Projects

17

Awards

4

SKT promp.T Top Prize · SSAFY Excellence · Oregon UAS · Startup Camp Best Team

Hackathons & challenges

9

Samsung SW-AI Academy for Youth

SSAFY 14th

### 02. Core Projects

Development projects built with AI coding agents and verified with reproduction tests, independent review, and before/after measurements come first. What the AI wrote and what I directed and verified are stated separately.

DEVELOPMENT EVIDENCE

**Core 3 Projects**

Ops Sentinel, JangBogo, and RE:RUN — organized as problem, role, technical choices, troubleshooting (reproduce → cause → fix → before/after), and limits.

## SKALA Cohort 4 Backend Final Lab (individual)

## Ops Sentinel | A Spring Boot API That Judges Metric Anomalies with a Rule Engine and Records Incidents, Actions, and Audit Logs

A backend where a rule engine sets severity and actions when virtual infrastructure metrics cross thresholds, a lock prevents duplicate incidents for the same resource, and decisions are recorded as AOP audit logs. The LLM (gpt-4o-mini) only summarizes decisions already made in one or two sentences.

### PROBLEM DEFINITION

The submitted README claimed '100% of decisions recorded in the audit log' and 'fixed by growing the connection pool from 30 to 60'. In a September 2026 re-measurement with a 3-second injected AI delay and 150 concurrent requests to the same resource, only 172/450 audit records (38.2%) survived, and requests that ended in 409 left neither a success nor a failure record.

### ROLE

Topic and scope, directing the AI coding agent and designing its verification, directing and reviewing the re-measurement and fixes (code written by Claude Code)

### KEY CONTRIBUTION

- Split the boundary so a deterministic rule engine decides and the LLM only summarizes afterwards; LLM output only goes into the aiSummary field.
- Pool 10, 150 concurrent requests to one resource, 3 s AI delay: before, 201 × 24 / 500 × 372 (p95 30.19 s) → after, 201 450/450 and audit 450/450 (p95 0.11 s). Re-measured Sep 2026, H2 in-memory, single JVM.
- 20 requests to one resource with a 3 s AI delay: p95 3.05 s at 6.6 rps → 0.05 s at 381.9 rps (pool 30).

### TECH STACK

Java 21

Spring Boot 3.3

JPA + MyBatis

Concurrency

JaCoCo

GitHub Actions

AI Coding Agent

### RESULT / LEARNING

Re-measured my own submitted claims, published the false ones (100% audit, fixed by pool growth, 3 s timeout) in the README, and then fixed the structure.

Google x Solana AI Agentic Hackathon

## JangBogo | A Merchant Gateway Where AI Agents Go from Quote Verification to Payment

An agent-commerce gateway: a buyer agent compares merchant quotes and pays on-chain, while a deterministic merchant-side policy engine verifies mandate signature, expiry, scope, cart, budget, and on-chain payment in six steps.

### PROBLEM DEFINITION

Even when agents can buy on their own, merchants are left asking whether they can trust an agent's order. Card payments need a human in the authentication loop, so an agent cannot complete checkout by itself.

### ROLE

Two-person entry (team Sagong-i); implementation and commits are mine | Directed the agent's plan→build→deploy→QA loop, set up Solana, Gemini, and GCP accounts and infrastructure, submitted, and reviewed the post-hoc check and fixes (code written by Claude Code)

### KEY CONTRIBUTION

- Unlike competing entries clustered on the buyer side, framed the problem on the merchant side (onboarding + verification).
- Designed the trust boundary so the LLM only writes the rationale, while a deterministic policy engine enforces budget, scope, and expiry.
- At submission, sequential tests confirmed five failure/attack types are blocked — over-budget, expired mandate, out-of-scope, underpayment, replay (11 unit + 20 integration = 31/31). Concurrent requests were never tested.

### TECH STACK

AI Agents

Gemini

Solana

x402

Next.js

Cloud Run

### RESULT / LEARNING

Separated what AI should do (explaining the choice) from what it must not (payment approval), and built that into the architecture.

Demo built for Wanted AI Championship 2026

## RE:RUN | When a Workflow Fails, AI Proposes a Fix and It Only Re-runs After a Human Approves

A Next.js demo: when a résumé-extraction workflow fails its data contract (zod), a real LLM (gpt-4.1-mini) proposes a root cause and a prompt fix, and the workflow re-runs from the failed step only after a human reviews the diff and approves. The server enforces the approval gate with HMAC signatures.

### PROBLEM DEFINITION

'AI fixes itself' demos are easy to overstate. In this one, the diagnosis prompt actually contained the answer's direction ('compute years of experience from employment periods as of now'), and the approval gate lived only in the UI, so calling the server directly bypassed it.

### ROLE

Product principles and acceptance criteria, measurement design, directing and reviewing fixes (MVP and fix code written by Claude Code)

### KEY CONTRIBUTION

- Before any code, had CONTRACT.md set acceptance criteria A1-A8 and a scope guard: 'any code path that bypasses the approval gate is a defect'.
- Hint-removal experiment: correct self-repair 72% → 0% (n=25 per condition, success criteria pre-registered, re-measured Sep 28, 2026).
- A loose check (!r.ok) reported '10/10 deterministic failures' even against an empty port with no server → a strict check (HTTP 200 + error text + raw null) gives 0/10 on the empty port and 20/20 sequential + 10/10 concurrent on the real server.

### TECH STACK

Next.js 16

TypeScript

OpenAI gpt-4.1-mini

zod

Human-in-the-loop

vitest

Playwright

### RESULT / LEARNING

Confirmed a result unfavorable to my own demo (72% → 0%) with a pre-registered experiment and published it on the README's first screen.

### SUPPORTING / HACKATHON

## Supporting Projects & Hackathons

Additional projects are separated as archive cases that show domain expansion, rapid execution, and supporting ideas.

COMPLETED

Sep 2 - 4, 2026 (3 days)

SKALA Cohort 4 Mini-project / AI Web Service Design

## ARGUS | An On-Premise AI Agent for Semiconductor Fab Part-Replacement Approval

An on-premise approval workflow for semiconductor fab part replacements: AI agents review spec compatibility, applicable regulations, and safety documents and attach the evidence, while the safety manager makes the approval decision.

AI Agents

On-Premise

Human-in-the-loop

FastAPI

Vue 3

### ROLE

Team: backend (Spring Boot), final presentation, bridge between lead and members | Personal re-implementation: directing, reviewing, and integrating contract-driven agent tracks (code written by Claude agents)

COMPLETED

Aug 2026

2026 Jeonnam-Gwangju Youth AI Solvathon

## Mudeungsan A!Job Radar | AI Recommendations Connecting Youth to Local Jobs

A service plan tackling the gap between existing jobs and the youth who never see them, using a structured-extraction + Korean sentence-embedding + generative-AI pipeline; it passed document screening and advanced to the finals.

Problem framing

AI recommendations

Vector search

Competitive analysis

Local jobs

### ROLE

Team lead · PM · Frontend lead | Planning, PM & pitch, React frontend, LLM/prompt recommendations & vector search

COMPLETED

Aug 2026 (awarded Aug 20, 2026)

SKT 'prompt.T for Everyone' Contest · Life AX

## Ansim Call | An AI That Doesn't Judge for You — A Voice-Phishing Companion for Seniors

A solo entry planning 'Ansim Call', an AI companion that helps seniors judge voice-phishing attempts — winner of the top prize (Life AX category) in SKT's 'prompt.T for Everyone' contest.

AI service planning

Seniors

Voice phishing

Life AX

### ROLE

Solo | Problem framing, service planning

COMPLETED

Jul 3 - 5, 2026 (72 hours)

Defense Tech / Hackathon

## SUPPORTY | AI-Copilot Operator Console for Contested Logistics

A contested-logistics control system built in 72 hours at the D4D (Deploy for Defense) APAC Seoul hackathon, where a single operator commands ground (UGV) and aerial (VTOL) unmanned assets alongside an AI copilot.

Defense Tech

AI Copilot

NL Command

Unmanned Systems

### ROLE

Planning & Frontend | Operator console design, demo scenario design, console & landing page frontend

COMPLETED

Jun 24 - 26, 2026 (3 days)

SSAFY Startup Camp

## EasyExam | An AI SaaS Turning Academy Exam Work into One Loop

An AI question-bank, exam-builder, and auto-grading SaaS for academy instructors — built from idea to live service in 3 days at the SSAFY Startup Camp (Samsung · JA Korea). Selected as Best Team.

EdTech SaaS

AI Pipeline

Business Model

Startup

### ROLE

Marketing & SEO | Market-customer validation and growth design



COMPLETED

Mar 2026 - May 2026

SSAFY 14th autonomous project final presentation

## Promptree | Marketplace for Executable AI Know-how

A marketplace that packages expert AI know-how as executable workflow products users can run immediately.

Next.js

FastAPI

AI Marketplace

Prompt Engineering

GA4

Fabric.js

### ROLE

AI model integration support, marketing, business/service planning support, presentation and market/competitor analysis contribution, Q&A preparation

COMPLETED

May 16, 2026

AI / Hackathon / Prototype

## GenWing Live | AI-Driven Event Networking App

An AI networking web app prototype built using Genspark OpenClaw at the Genspark Meetup & Hackathon.

Genspark OpenClaw

AI Agent

Prototype

PWA

### ROLE

Solo planning & development | Problem framing, UX design, agent workflow, and deployment

COMPLETED

Feb 2026 - Apr 2026

SSAFY specialized project

## DailyLog | AI-Driven Reflection & Decision Support

A service that solves user retention issues by defining the problem as 'input friction' and 'broken reflection flows', using AI-driven questioning.

AI reflection

Recommendation

Product planning

PM

### ROLE

PM & Frontend | Problem framing, user journey design, prioritization, UI implementation, and presentation

COMPLETED

Mar 28, 2026 (Round 1 · Day 1)

AI R&D / Hackathon

## MultiplierBoard | Binary Multiplication with 1-Layer Attention (KRAFTON AI R&D)

A solo Round-1 task from the KRAFTON AI R&D Hackathon: designing and verifying two solutions — a constructive-exact model and a learned model — for 6-bit binary multiplication under a 1-layer causal self-attention constraint, within a single day.

Transformer

AI Pair Workflow

Verification

AI R&D

### ROLE

Solo | Problem analysis, solution direction, generative-AI execution & verification workflow, result verification and reporting

COMPLETED

Jan 2026 - Feb 2026

SSAFY project

## Loggy | Decision Recording & Collaboration Platform

A platform that maximizes collaboration efficiency by formalizing scattered discussions into Discussion, Approval, and Record structures.

Collaboration platform

GitHub workflow

Real-time

PM

### ROLE

Planning & Frontend Lead | Service planning, decision flow structuring, real-time UI & state architecture, and frontend lead

COMPLETED

Feb 2026

Jeju AWS Global Space Challenge hackathon

## Tamna-AI | Distributed Data Center Location Analysis

A service that reframes Jeju's energy curtailment as a grid mismatch issue and analyzes optimal locations for data centers to absorb surplus power.

Energy domain

Location analysis

Business planning

PM

### ROLE

Planning | Problem reframing, business model design, and service structuring

COMPLETED

Dec 2025

SSAFY integrated project

## S.A.L.E.S | Customer-Centered CRM Project

A CRM project that structured fragmented lead management into a '5-stage customer journey' and 'Kanban-style pipeline'.

CRM

Customer Journey

Business Planning

PM

### ROLE

Planning | Defining 5-stage journey, pipeline UI design, mapping business logic to data models

COMPLETED

Sep 2025

Jeonju ICT Coding Challenge

## Jeonju ICT Coding Challenge | Regional Issue-Solving Service

A project defining regional tourism, social, and environmental issues and materializing them into actionable service features and implementation.

Regional issues

Public service

Planning & Implementation

### ROLE

Problem framing, service planning, and implementation support

COMPLETED

Aug 2025

Shinhan Bank X SSAFY hackathon

## Campus Chronicle | Growth Data-Driven Financial Service

A platform where students link learning and activity data to financial value, proving trust based on their growth process.

Fintech

Growth data

Product planning

Financial UX

### ROLE

UI/UX Design & Frontend | End-to-end app/game UI/UX in Figma, design system & components, React Native sub-screens (quests, chronicle feed)

### 03. AI in Action

Real cases where AI was used as a problem-solving tool, not decoration. Each case follows problem → how AI was used → outcome.

JailyLog

#### A reflection flow where AI asks first

##### PROBLEM

Users gave up journaling because they faced a blank screen with no idea what to write.

##### HOW I USED AI

Structured each day into Event · Emotion · Cause slots and designed a prompt flow where AI asks context-aware questions first.

##### OUTCOME

Became the core feature that lowered the barrier to reflection, leading to the SSAFY Specialized Project Excellence Award (3rd place).

JailyLog

#### Turning records into action recommendations

##### PROBLEM

Accumulated records ended up as read-only data and never drove behavior change.

##### HOW I USED AI

Balanced exploration and exploitation with Thompson Sampling, and used pgvector search to bring past reflections into the current context.

##### OUTCOME

Built and shipped an AI data flow that connects recording → understanding → recommendation.

SUPPORTY

## A command UX that translates natural language into control sequences

### PROBLEM

A single operator had to control ground (UGV) and aerial (VTOL) assets at once — sticks and screen-switching alone caused cognitive overload.

### HOW I USED AI

Designed a command UX translating orders like 'forward 5 seconds' into control sequences, plus an AI triage queue (life criticality → time sensitivity → mission impact) and a voice copilot.

### OUTCOME

Demoed a live console wired to a real UGV & VTOL within 72 hours, leading to the Oregon UAS Accelerator award at D4D Seoul.

Promptree

## Making prompts a sellable product

### PROBLEM

AI know-how (prompts) loses its value the moment it is copy-pasted, so no market could form.

### HOW I USED AI

Structured prompts into slot-based executable templates and improved the execution flow based on GA4 drop-off data.

### OUTCOME

Extended the idea into an 'AI know-how market' business model with market and competitor analysis.

JenWing

## AI agents as execution partners

### PROBLEM

A hackathon deadline demanded a working product, not just an idea.

### HOW I USED AI

Fed structured UX designs and requirements to the Genspark OpenClaw agent, completing a solo planning-to-development cycle.

### OUTCOME

Shipped an AI networking PWA prototype within the event.

## 04. Core Competencies

How I verify the core projects built with AI coding agents, plus the planning, AI/Data, collaboration, and communication skills from earlier team projects — each tied to evidence.

J1

### Measurement- and Verification-Driven Development

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*I check code written by AI coding agents before trusting it: defects start with a reproduction, claims come with their measurement conditions.*

#### Reproduction Test First

For each defect, a reproduction test was committed first and its failure confirmed before the fix. Ops Sentinel (AI call inside the lock: 10 requests at 1,656-1,670 ms → under 750 ms), JangBogo (payment race reproduced as 6 failures → passing).

#### Review Separated from Authoring

A reviewer separate from the authoring agent checked the work. It caught a payment regression introduced by the fix in JangBogo, a second approval-gate bypass in RE:RUN, and async summary overwrites in Ops Sentinel, all before merge.

#### Before/After with Conditions

Numbers come with their conditions. Ops Sentinel pool 10, 150 concurrent, 3 s AI delay: audit 450/450 (500 × 372 before); RE:RUN self-repair 72% → 0% without the hint (n=25); JangBogo max orders 10 → 1.

#### Publishing Disproved Claims

Claims that re-measurement proved wrong (Ops Sentinel '100% audit', RE:RUN 'AI self-diagnosis' with the prompt hint left unstated) were corrected in the README, not deleted. What AI wrote and what I did are stated separately.

J2

## Product & Planning

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*How I reframed problems and translated them into service structures in DailyLog, Loggy, and Promtree.*

### **Problem Framing**

Reframed journaling failure, lost collaboration context, and AI know-how exchange as user behavior and service-structure problems.

### **User Flow**

Designed question flows, commit flows, and slot-based execution flows so users could move from uncertainty to the next action.

### **PRD / Policy and Service Structure**

Organized key features, information structure, roles, and priorities into documents and screen flows so teams could implement against shared criteria.

### **Market / Competitor Analysis**

In Promtree, reviewed AI know-how commercialization, revenue models, and competitor positioning to test business viability.

J3

## AI & Data

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*I connect AI to service layers that support user judgment and action, rather than treating it as decoration.*

### **LLM / Prompt Engineering**

Handled prompts as design elements that shape experience and output quality, as in DailyLog's question flow and Promtree's slot execution.

### **Python / SQL**

Through SSAFY training in Python, SQL, AI, and Data, I consider data and implementation constraints during planning.

### **Data-Informed Decisions**

Reviewed how user behavior data can inform product decisions through recommendation, retention, and collaboration-record structures.

### **AI Feature Planning**

Designed AI features to support reflection, recording, and execution while preserving user control instead of replacing user judgment.

J4

## Web & Collaboration

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*Execution skills for turning planning intent into screens, data flow, and team workflows.*

### **React Implementation Literacy**

Participated in frontend implementation and checked how screen flow and state changes affect user experience.

### **Git / GitHub**

In Loggy, interpreted Git's PR/Merge structure as a collaboration decision UX and adapted it into a service model.

### **Notion / Jira / Slack / Mattermost**

Used collaboration tools to organize schedules, roles, meeting notes, and presentation materials around shared execution criteria.

### **Collaborative Documentation**

Documented feature definitions and decision rationale so teammates could implement and present from the same context.

J5

## Communication & Leadership

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*Communication skills built from philosophical reasoning and SSAFY team-project practice.*

### **Presentation**

Focus on explaining project problems, solution structures, and lessons learned in a way interviewers and users can understand.

### **Team Alignment**

Helped synthesize team ideas into one goal and narrow them into implementable priorities.

### **User-Centered Explanation**

Use question framing and logical explanation from Philosophy training to explain why users drop off and what they need.

### **Feedback Loop**

Adjusted project direction by reflecting feedback from presentations, meetings, and implementation into screen flows and documentation.

## 05. Selected Experience



## **AI Training & Team Projects**

### **SKALA (SK AI Leader Academy), Cohort 4**

**2026 - PRESENT**

- Currently enrolled in SKALA (SK AI Leader Academy) Cohort 4, working on team projects including AI web service design.  
Led backend design and the final presentation for the ARGUS mini-project (five-person team).

## **AI/SW Training & Product Projects**

### **Samsung SW·AI Academy For Youth, SSAFY**

**2025.07 - 2026.06**

- Learning service development and AI/SW technical literacy based on Python, SQL, Web, AI, and Data.
- Worked on problem framing, feature prioritization, screen-flow organization, and presentation materials in team projects including DailyLog, Loggy, and Promtree.
- Helped share notices and support communication among trainees as the class representative (CA) at the Gwangju campus.

## **Core Project Practice**

### **DailyLog / Loggy / Promtree**

**2026.01 - 2026.05**

- In DailyLog, reframed journaling failure as input friction and reflection-flow problems, then connected it to AI questioning and recommendation structures.
- In Loggy, organized disappearing collaboration context into a recording flow inspired by Git's PR/Merge structure.
- In Promtree, structured AI know-how as an executable workflow product while reviewing market viability, revenue models, and competitor positioning.

## **B.A. in Philosophy**

### **Chonnam National University**

**2016.03 - 2022.02**

- Studied philosophy and trained by reading texts, organizing questions, and explaining ideas logically.
- Later connected SW·AI learning and service planning experience into an interest in problems between people and technology.

## **Early Global & Leadership Experience**

### **Portage High School / Gwangju Goryeo High School**

**2012.01 - 2016.02**

- Lived for one year as an exchange student at Portage High School in Indiana.
- Served as student council president in high school, gaining experience speaking in front of others and coordinating opinions.

## **Military Service**

### **Republic of Korea Marine Corps**

**2016.12 - 2018.09**

- Served as a rifleman in the 2nd Marine Division.
- Partly supported interpretation and situation updates during ROK-US joint exercises.  
Learned the habit of carrying assigned work through in constrained environments.

## **Customer Experience Operation**

### **Charancha / Jaguar Land Rover Gangdong Service Center**

**2022.12 - 2024.01**

- Handled valet operations and customer service at Jaguar Land Rover Gangdong Service Center.
- Managed vehicle movement, customer waiting, and on-site communication while experiencing the flow of a service operation.  
Observed firsthand where customers tend to feel friction in the service process.

## **Sales & Field Operation**

### **Yeyoung Design**

**2022.08 - 2022.11**

- Handled interior film sales and customer consultation.
- Supported on-site installation work and experienced the gap between customer requests and actual field conditions.  
Learned how requirements explained in conversation can change when they meet the real work environment.

## **06. CONTACT**

# **Let's Shape the Future Together**

I am open to conversations about AI/SW service planning, PM roles, portfolios, and project collaboration.

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**VISIT GITHUB   LINKEDIN   SEND EMAIL   VIEW PORTFOLIO PDF**

Built with React and an AI/SW-focused portfolio structure.

Designed to present problem framing, service planning, and AI-enabled project experience.